

| Calculations                                                         |           |           |           |           |
|----------------------------------------------------------------------|-----------|-----------|-----------|-----------|
|                                                                      | Sheep     | Pig       | Cow       | Goat      |
| Meat Diet Portion                                                    | 0.71%     | 7.26%     | 9.47%     | 2.37%     |
| Meat MCal Target                                                     | 3,891.74  | 39,730.43 | 51,828.43 | 12,954.40 |
| Potential Dairy Value attached to each juvenile                      | 132.54    | 0.00      | 1015.20   | 126.90    |
| Value per juvenile                                                   | 48.00     | 160.00    | 640.00    | 48.00     |
| Adjusted Dairy Value attached to juvenile based on Dairy Utilization | 66.27     | 0.00      | 507.60    | 63.45     |
| Culled Mothers Value                                                 | 14.77     | 4.41      | 105.26    | 14.77     |
| Culled Adult Males                                                   | 0.74      | 0.22      | 5.26      | 0.74      |
| Final Value per juvenile                                             | 63.51     | 164.63    | 750.53    | 63.51     |
| Value MCals                                                          | 63.51     | 164.63    | 750.53    | 63.51     |
| Offspring required Year <sup>-1</sup>                                | 61.28     | 241.34    | 69.06     | 203.98    |
| MCals from meat                                                      | 3891.74   | 39730.43  | 51828.43  | 12954.40  |
| Max Utilized Mcals allowed from Dairy                                | 4061.01   | 0.00      | 35052.88  | 12942.63  |
| MCals actually Utilized from Dairy                                   | 4061.01   | 0.00      | 35052.88  | 12942.63  |
| Percent of diet from their meat                                      | 0.71%     | 7.26%     | 9.47%     | 2.37%     |
| Percent of Diet from their Dairy                                     | 0.74%     | 0.00%     | 6.40%     | 2.36%     |
| Kg Fleece                                                            | 464.57    | 0.00      | 0.00      | 0.00      |
| 1 Year+ (age) skins                                                  | 23.07     | 9.47      | 13.44     | 76.09     |
| Yearly Offspring skins                                               | 112.09    | 284.91    | 102.07    | 371.70    |
| Birthing Events per year                                             | 1.00      | 1.86      | 1.00      | 1.00      |
| Offspring Mother <sup>-1</sup> Year <sup>-1</sup>                    | 0.56      | 5.48      | 0.68      | 0.56      |
| Mothers Step 1                                                       | 109.43    | 44.08     | 101.55    | 364.25    |
| Males Step 1                                                         | 30.64     | 120.67    | 34.53     | 101.99    |
| Females Step 1                                                       | 30.64     | 120.67    | 34.53     | 101.99    |
| Replacement Mothers per year                                         | 22.19     | 9.12      | 12.88     | 73.15     |
| Breeding Males Required                                              | 5.74      | 2.31      | 5.33      | 19.12     |
| Additional Mothers                                                   | 90.72     | 7.96      | 48.55     | 299.50    |
| Males Step 2                                                         | 25.40     | 21.79     | 16.51     | 83.86     |
| Females 2                                                            | 25.40     | 21.79     | 16.51     | 83.86     |
| Total Mothers                                                        | 131.61    | 53.20     | 114.43    | 437.40    |
| Total Males                                                          | 56.04     | 142.45    | 51.04     | 185.85    |
| Total Females                                                        | 5.24      | 98.88     | 18.02     | 18.13     |
| Total Offspring                                                      | 112.09    | 284.91    | 102.07    | 371.70    |
| Total Animals                                                        | 198.64    | 296.85    | 188.82    | 660.51    |
| People per person                                                    | 1.00      |           |           |           |
| Total Adults per Hectare                                             | 1.45      | 0.23      | 0.36      | 1.45      |
| Total Offspring per Hectare                                          | 0.71      | 1.00      | 0.22      | 0.71      |
|                                                                      | 0.49%     |           | 5.97%     | 1.63%     |
| Total Adults                                                         | 228       | 63        | 168       | 756       |
|                                                                      |           |           |           |           |
|                                                                      |           |           |           |           |
| Total Offspring                                                      | 112       | 285       | 102       | 372       |
| Days for Maintenance                                                 | 5.00      | 220.00    | 0.00      | 5.00      |
| Dry Mothers                                                          | 19.53     | 0.00      | 12.36     | 65.70     |
| Other                                                                | 1,793.37  | 0.00      | 0.00      | 5,947.21  |
| MCals Dry Mothers                                                    | 22,809.17 | 0.00      | 26,175.57 | 76,740.64 |
| total Maintenance MCals                                              | 24,602.54 | 0.00      | 26,175.57 | 82,687.85 |

| MCals Required |           |           |           |           |
|----------------|-----------|-----------|-----------|-----------|
|                | Sheep     | Pig       | Cow       | Goat      |
| Gestating      | 66,747    | 130,075   | 170,866   | 221,348   |
| Lactating      | 127,340   | 46,241    | 183,727   | 422,289   |
| Maintenance    | 24,603    | 0         | 26,176    | 82,688    |
| Adult Males    | 6,710     | 0         | 11,287    | 22,336    |
| Offspring      | 30,767    | 286,334   | 373,577   | 102,032   |
| TOTAL          | 256,167.4 | 462,650.0 | 765,631.8 | 850,692.6 |

| Animal Diet Breakdown           |        |        |        |        |
|---------------------------------|--------|--------|--------|--------|
|                                 | Sheep  | Pig    | Cow    | Goat   |
| Crop                            | Barley | Barley | Barley | Barley |
| Grams <sup>-1</sup> Straw/Chaff | 1000.0 | 0.0    | 4956.0 | 1000.0 |
| MCal kg <sup>-1</sup>           |        |        |        |        |
| Grams <sup>-1</sup> Grain       | 750.0  | 750.0  | 1500.0 | 750.0  |
| MCal kg <sup>-1</sup>           |        |        |        |        |
| Days                            | 0.0    | 0.0    | 0.0    | 0.0    |
| Number of Animals               | 112.1  | 284.9  | 102.1  | 371.7  |
| MCals from Straw/Chaff          | 0.0    | 0.0    | 0.0    | 0.0    |
| MCals from Grain                | 0.0    | 0.0    | 0.0    | 0.0    |
| MCals from other feed           | 0.0    | 0.0    | 0.0    | 0.0    |

| Diet Calculations                           |                  |         |
|---------------------------------------------|------------------|---------|
| Plants Mcal                                 | 385,943.474      |         |
| Meat Mcal                                   | 109,500          |         |
| Dairy                                       | 52,056.525779962 |         |
|                                             | Perfect          |         |
| Plants %                                    | 70.49%           |         |
| Meat %                                      | 20.00%           |         |
| Dairy %                                     | 9.51%            |         |
| check                                       | Perfect          |         |
| Percent                                     |                  |         |
| Domestic Meat                               | 19.80%           |         |
| Dairy                                       | 9.51%            |         |
| Wild Meat                                   | 0.20%            |         |
| Crops                                       | 63.44%           |         |
| Wild Plants                                 | 7.05%            |         |
| check                                       | Perfect          |         |
|                                             |                  |         |
| new dairy                                   | 52056.53         |         |
| potential dairy < limit                     | TRUE             |         |
| new limit                                   | 80.0%            |         |
| the limit greater than that left to satisfy | TRUE             |         |
| limit                                       | 1                |         |
| limit dairy checked                         | FALSE            |         |
| Mcalories                                   |                  | Surplus |
| Domestic Meat                               | 108,405.000      |         |
| Dairy                                       | 52,056.526       |         |
| Wild Meat                                   | 1095.000         |         |
| Crops                                       | 347,349.127      |         |
| Wild Plants                                 | 38,594.347       |         |
| check                                       | Perfect          |         |
|                                             |                  |         |
| SURPLUS DAIRY                               | 0.000            |         |
| This could account for additional % of diet | 0.00%            |         |

Herd Modelling

|                                                               | Sheep  | Pig    | Cow    | Goat   |  |  |  |  |
|---------------------------------------------------------------|--------|--------|--------|--------|--|--|--|--|
| Mothers Req'd<br>Males for Slaughter<br>Females for Slaughter | 132    | 53     | 114    | 437    |  |  |  |  |
|                                                               | 56     | 142    | 51     | 186    |  |  |  |  |
|                                                               | 5      | 99     | 18     | 18     |  |  |  |  |
|                                                               |        |        |        |        |  |  |  |  |
| Offspring available for Consumption                           | 112    | 285    | 102    | 372    |  |  |  |  |
| Gestation Req'd                                               | 36492  | 110182 | 115600 | 121471 |  |  |  |  |
| Lactation Req'd                                               | 69620  | 39169  | 124301 | 231743 |  |  |  |  |
| Maintenance Req'd                                             | 83131  | 0      | 34386  | 275900 |  |  |  |  |
| Adult Males                                                   | 6710   | 0      | 11287  | 22336  |  |  |  |  |
| Offspring Required                                            | 36820  | 311976 | 447068 | 122104 |  |  |  |  |
| Energy Required                                               | 232773 | 461327 | 732641 | 773554 |  |  |  |  |
| Ewe Fleece Kg Yr-1                                            | 296    |        |        |        |  |  |  |  |
| Ram Fleece Kg year <sup>-1</sup>                              | 12     |        |        |        |  |  |  |  |
| Juvenile Fleece Kg Yr-1                                       | 157    |        |        |        |  |  |  |  |
| Kg Fleece Year-1                                              | 465    | 0      | 0      | 0      |  |  |  |  |